VAVEhub · VE Academy capstone

Value study: K-17 cordless kettle

Prepared by Priya Shah · 15 October 2026

£9.00	£7.88	£7.85
starting cost	target	after approved ideas

1 · SCOPE AND STARTING POINT

In scope: body, lid, controls, base, trim, assembly. Out of scope: heating element contract and mains cable.

Why this target: —

Biggest cost items: Heating element, Thermostat and switch, Body (moulded, coloured)

Requirement questioned: —

2 · FUNCTIONS, COST AND WORTH

Basic function: Heat water · **Others:** Control temperature, Contain water

FUNCTION	COST	WORTH	VI	TARGET?
Heat water	£2.10	£2.00	1.05	
Contain water	£1.70	£0.90	1.89	
Control temperature	£1.30	£0.60	2.17	✓
Conduct current	£1.20	£0.70	1.71	
Convey esteem	£1.30	£0.30	4.33	✓

3 · IDEAS

14 ideas generated, including: Convey esteem: moulded-in colour and an in-mould film instead of the chrome trim; Convey esteem: metallic-effect plastic, as used on car interior trim; Convey esteem: drop the trim, change the handle colour instead; Control temperature: one combined thermostat and switch from a catalogue; Control temperature: bimetal disc instead of the electronic board; ...

4 · EVALUATION (SAVING 50% · RISK 30% · EFFORT 20%)

IDEA	SAVING PER KETTLE	S / R / E	SCORE
Convey esteem: moulded-in colour and an in-mould film instead of the chrome trim	£0.62	7 / 8 / 7	7.3
Control temperature: one combined thermostat and switch from a catalogue	£0.38	6 / 7 / 6	6.3
Assemble: snap-fit base cover, three screws instead of seven	£0.15	4 / 8 / 8	6.0
Contain water: filter moulded into the lid	£0.08	3 / 7 / 7	5.0

5 · BUSINESS CASE: CONVEY ESTEEM: MOULDED-IN COLOUR AND AN IN-MOULD FILM INSTEAD OF THE CHROME TRIM

£155,000
saving a year

£59,200
one-time cost

4.6 mo
payback

Rig test
highest test

Risk and how it's handled: —

6 · DECISIONS

IDEA	DECISION	OWNER	BY	STAGE
Convey esteem: moulded-in colour and an in-mould film instead of the chrome trim	Approve	Sam	Mar 2027	L3
Control temperature: one combined thermostat and switch from a catalogue	Approve with conditions	Ana	May 2027	L2
Assemble: snap-fit base cover, three screws instead of seven	Approve	Lee	Feb 2027	L3
Contain water: filter moulded into the lid	Needs more data	Sam	Jan 2027	L1

> The approved ideas take the K-17 from £9.00 to £7.85, inside the £7.88 target: £287,500 a year at 250,000 kettles. Savings will be reported only when each change is in production and checked by finance against the £9.00 starting cost.